Public repository copy – filtered, not the complete submission draft. Tissue images, individual-slide plots, case-level tables and row-level source data are withheld. Marked omissions preserve the original numbering and captions. References to Source Data describe the local research archive; only screened aggregate tables are distributed here. Reported aggregate estimates are unchanged.

Foundation-model-based whole-slide classification of salivary gland MALT lymphoma and Sjögren's syndrome

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Abstract

Background: The histological distinction between salivary gland mucosa-associated lymphoid tissue (MALT) lymphoma and Sjögren's syndrome (SS) motivates disease-specific evaluation of computational pathology. We examined whether frozen foundation-model features support whole-slide classification and how model choice and sampling affect performance.

Methods: The internal dataset contained 193 slides, divided into 144 training and 49 validation slides used for checkpoint selection. Historical external evaluation included 168 slides (152 MALT, 16 SS) with incompletely confirmed stain categories. We evaluated a retained Hybrid-Evidence multiple-instance learning model, simpler baselines, controlled training across seeds and repeated patch sampling.

Results: The retained model achieved validation AUROC 0.9599 (95% CI, 0.9046–0.9967) and external AUROC 0.8721 (0.7364–0.9758). External sensitivity was 93.42%, specificity 68.75% and negative predictive value 52.38% at threshold 0.5. Paired AUROC differences from ABMIL included zero. Controlled five-seed external means were 0.8056 for Hybrid and 0.8061 for internally selected Attention+L2, which used 56.75% fewer parameters. Linear controls achieved external AUROCs of 0.8335–0.8462 on common reference inputs. With the retained checkpoint, 18/168 external slides crossed the decision threshold across 20 patch samples.

Conclusions: Frozen pathology features contained discriminatory information for the evaluated MALT/SS categories, but controlled comparisons did not establish an advantage for the more complex aggregator. Clinical interpretation requires a defined material spectrum and operating point, completed patient-level documentation and independent evaluation.

Keywords: MALT lymphoma; Sjögren's syndrome; salivary gland; computational pathology; multiple-instance learning

1 Introduction

Distinguishing salivary gland mucosa-associated lymphoid tissue (MALT) lymphoma from lymphoid-rich inflammation associated with Sjögren's syndrome (SS) requires assessment of the infiltrate in its glandular context. SS is an autoimmune setting in which MALT lymphoma can arise, and lymphoepithelial lesions or architectural disturbance alone do not resolve every differential diagnosis [1, 2]. Integrated assessment may require morphology, immunophenotyping and molecular findings within the relevant lymphoma classification framework [3, 4]. A computational analysis of routine whole-slide images could provide a quantitative view of this material. The immediate research question is whether slide images contain a reproducible classification signal for the two recorded diagnostic categories.

Disease-focused digital pathology has begun to address related questions. Cell–tissue graph models have been used for SS classification, and an attention model evaluated whether minor labial salivary

gland biopsies had a focus score of at least one [5, 6]. Other studies classified orbital MALT lymphoma versus IgG4-related ophthalmic disease, or distinguished MALT and diffuse large B-cell lymphoma categories using pathologist-selected tissue images [7, 8]. These studies establish relevant approaches, but their organs, reference outcomes or image-selection procedures differ. They do not directly answer how whole-slide, slide-label-supervised models behave for salivary gland MALT versus SS.

Multiple-instance learning (MIL) connects slide-level labels to collections of image patches without requiring exhaustive lesion annotations [9–11]. Pathology foundation models now supply transferable patch representations, allowing downstream classifiers to be trained while keeping the image encoder fixed [12, 13]. This makes disease-specific experiments feasible with limited slide collections, but leaves an important empirical question: how much does a complex aggregator add beyond simpler summaries of the same features? Recent work on MIL training and feature sampling motivates separating architectural complexity from optimization and input-sampling effects [14].

We studied retrospective MALT–SS classification using fixed UNI2-h representations and Hybrid-Evidence MIL. The analysis combines a retained model's internal validation and external performance with simpler baselines, controlled retraining and repeated patch sampling. We also examined how external archive categories related to scores and made selected image regions available for pathological review. The study evaluates slide classification and the conditions under which its measured performance changes; it does not treat a slide score as a replacement for integrated clinical diagnosis.

2 Results

2.1 Study material and evaluation framework

The development dataset comprised 193 whole-slide images (WSIs): 90 MALT and 103 SS. A stratified split assigned 144 slides to training and 49 to internal validation (Fig. 1; Table 1). The validation set was used to select checkpoints; its performance estimates therefore describe model development. The external collection contained 227 slides, of which 223 yielded predictions and 168 entered the author-reviewed analysis set (152 MALT and 16 SS). Patient characteristics and the clinical reference-standard description are reserved for the hospital records in Table 1 and Methods.

Each slide was represented by frozen UNI2-h patch features. The principal Hybrid-Evidence multiple-instance learning (MIL) model combined attention pooling, feature-distribution summaries and latent-prototype summaries. We first evaluated its retained checkpoint, then compared simpler models and controlled retraining on the same data. These analyses distinguish the performance of one fitted model from the reproducibility of a training procedure.

2.2 The retained model discriminated MALT from SS

The retained Hybrid checkpoint achieved an internal validation area under the receiver operating characteristic curve (AUROC) of 0.9599 (95% confidence interval [CI], 0.9046–0.9967). At a score threshold of 0.5, it classified 22/23 MALT and 21/26 SS slides correctly (Fig. 2). External AUROC was 0.8721 (95% CI, 0.7364–0.9758), with 142/152 MALT and 11/16 SS slides classified correctly. External sensitivity was 93.42% and specificity was 68.75%; the specificity interval was wide (44.40%–85.84%; Table 2).

ABMIL achieved internal AUROC 0.9498 and external AUROC 0.7903. The paired Hybrid-minus-ABMIL differences were 0.0100 (95% CI, −0.0184 to 0.0452) internally and 0.0818 (−0.0136 to 0.2011) externally. External comparisons used identical sampled inputs. Both intervals included zero. These were comparisons of complete implementations with different representation dimensions and training

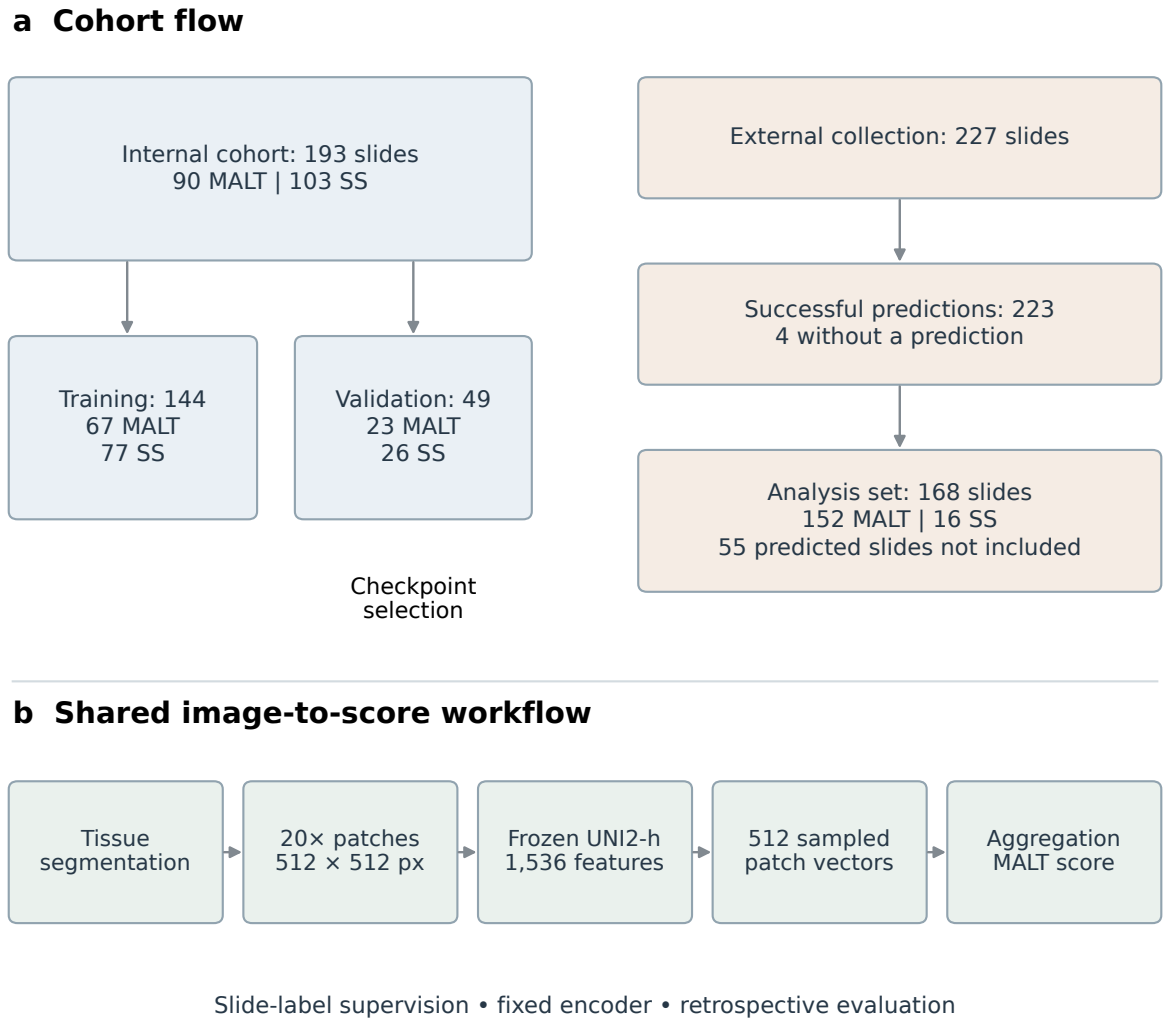


Figure 1: Cohort flow and evaluation framework. All counts are WSIs. The 49-slide validation set was used for checkpoint selection. The external flow distinguishes collection, successful inference and final analysis; clinical exclusion and failure reasons remain completion item H03. The later controlled training and algorithm analyses reused these data and did not create new independent clinical cohorts. The bottom panel summarizes the shared image-to-score workflow; the full Hybrid architecture is detailed in Supplementary Methods.

Table 1: Study cohorts and clinical fields reserved for source records

Characteristic	Training	Internal validation	External
Slides, n	144	49	168
MALT / SS, n	67 / 77	23 / 26	152 / 16
Role	Training	Checkpoint selection	External evaluation
Patients, n	— [H01]	— [H01]	— [H01]
Slides per patient	— [H01]	— [H01]	— [H01]
Age and sex	— [H01]	— [H01]	— [H01]
Gland site; specimen type	— [H01]	— [H01]	— [H01]
Collection dates; clinical source	— [H01]	— [H01]	— [H01]
Confirmed stain	— [H01]	— [H01]	— [H01]
Diagnostic reference standard	— [H02]	— [H02]	— [H02]

Dashes with completion codes are unfilled hospital-record fields, not zero counts or evidence that the records do not exist. Patient linkage and specimen-level attributes must be incorporated before a patient-level interpretation. All available counts refer to WSIs.

Table 2: Retained Hybrid performance with conditional 95% intervals

Metric	Internal validation	External evaluation
AUROC	0.9599 (0.9046–0.9967)	0.8721 (0.7364–0.9758)
AP	0.9514 (0.8864–0.9965)	0.9733 (0.9365–0.9971)
Sensitivity	95.65% (79.01%–99.23%)	93.42% (88.31%–96.39%)
Specificity	80.77% (62.12%–91.49%)	68.75% (44.40%–85.84%)
Accuracy	87.76% (75.76%–94.27%)	91.07% (85.79%–94.51%)
PPV	81.48% (63.30%–91.82%)	96.60% (92.29%–98.54%)
NPV	95.45% (78.20%–99.19%)	52.38% (32.37%–71.66%)
Balanced accuracy	88.21%	81.09%

Threshold metrics use 0.5. AUROC and AP intervals use 2,000 class-stratified WSI bootstrap draws; proportion intervals use Wilson's method. These intervals condition on fitted predictions and do not adjust for selection or patient clustering.

settings, rather than isolated tests of individual Hybrid components. The full historical model comparison, including the separate CONCH v1.5–TITAN pipeline, is provided in Supplementary Table S1.

2.3 External operating-point performance varied with material composition

The external MALT fraction was 90.48%. Overall accuracy of 91.07% was therefore close to the 90.48% accuracy of a constant MALT prediction, whereas balanced accuracy was 81.09%. Average precision (AP) was 0.9733. Of 21 negative model predictions, ten were MALT, giving a negative predictive value (NPV) of 52.38% (95% CI, 32.37%–71.66%). These quantities describe different aspects of performance in the observed case mix.

Archive categories showed different score distributions (Fig. 3). Among source-B MALT slides, 23/33 with H&E-like tags and 111/111 with immunohistochemistry (IHC)/special-stain tags were positive at threshold 0.5. All eight MALT slides in a separate HE directory were positive. These are archive-derived categories; the stain identity of the SS series was unspecified. In a post hoc descriptive analysis omitting the 111 IHC/special-stain-tagged slides, AUROC was 0.8689, sensitivity was 75.61% (31/41), and specificity remained 68.75% (11/16). Thus, the AUROC point estimate was

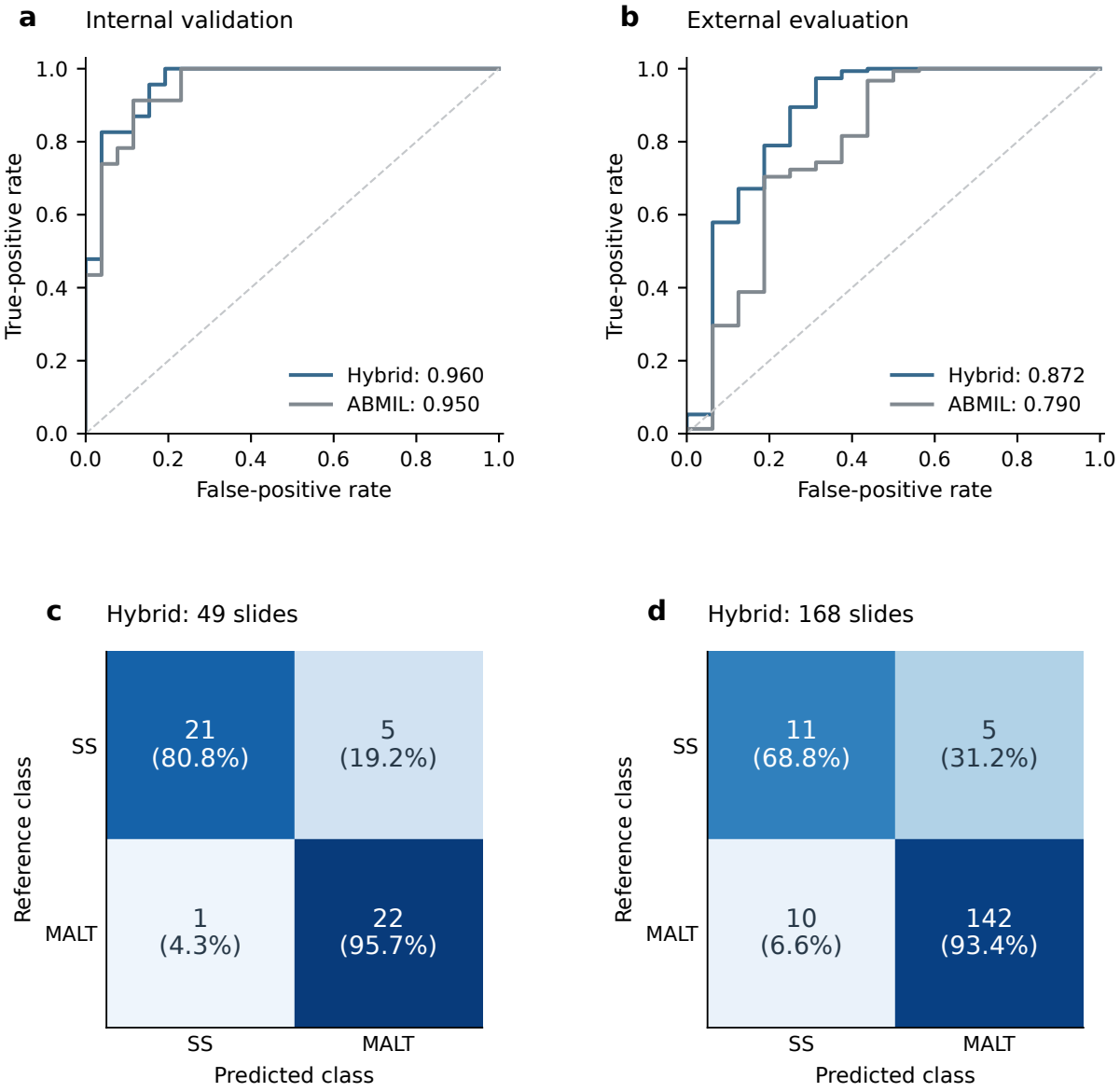


Figure 2: Discrimination and threshold performance of the retained checkpoint. a,b, ROC curves for Hybrid and ABMIL on internal validation (23 MALT, 26 SS) and external evaluation (152 MALT, 16 SS). External inputs were matched between models. AUROCs describe the retained complete implementations. **c,d,** Hybrid confusion matrices at threshold 0.5, showing counts and within-reference-class percentages. Validation was used for checkpoint selection. Conditional intervals appear in Table 2; paired AUROC-difference intervals included zero in both cohorts.

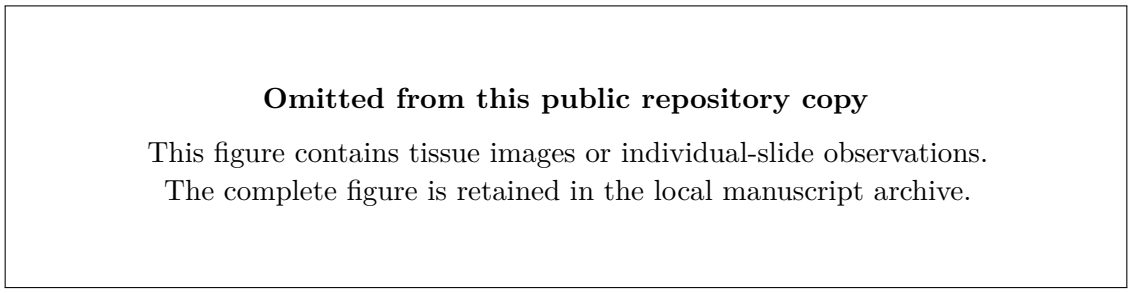


Figure 3: External scores in relation to archive categories. a, Each point is one slide; horizontal segments show medians and the dashed line marks 0.5. Labels give the reference class and sample count. IHC, immunohistochemistry. Categories were derived from archive names and directories, not pathology-confirmed staining. **b,** Descriptive metrics for all 168 external slides and the 57-slide subset after omitting the IHC/special-stain archive group. The subset contained 41 MALT and all 16 SS slides, whose stain remained unspecified; it is not an H&E-only validation set. The primary cohort and checkpoint were unchanged.

similar while the fixed-threshold MALT detection rate was lower. This subset does not constitute a pathology-confirmed H&E-only evaluation.

The external Brier score was 0.0919. For context, constant scores equal to the training MALT proportion and the observed external MALT proportion gave descriptive Brier scores of 0.2793 and 0.0862, respectively. The latter uses evaluation labels and is not a deployable comparator. We therefore interpreted the model's Brier score as squared prediction error, not evidence that its scores were calibrated clinical probabilities.

2.4 Controlled comparisons did not establish an aggregation advantage

Under a common training protocol, matched attention, full fixed scaling and full learned gating had internal AUROC means of 0.9348, 0.9398 and 0.9353 across three seeds. Their external means were 0.7270, 0.7445 and 0.7633, respectively (Supplementary Table S3). The learned-gate difference from matched attention changed direction across external runs. These results did not establish a consistent benefit from learned gating.

A subsequent screen completed 34 neural-network training runs and two deterministic linear controls. Eight modifications were tested alongside the original Hybrid and matched-attention controls; each configuration initially used three seeds. Attention with input L2 normalization was selected by internal mean AUROC, before the new external inference. Its internal mean of 0.9365 exceeded Hybrid's 0.9353 by 0.0011, below the predefined engineering target of 0.01. Two further seeds were then evaluated for the selected candidate and Hybrid, without changing the selection. Across all five seeds, internal means were 0.9341 and 0.9348, and external means were 0.8061 and 0.8056, respectively (Fig. 4; Table 3). The candidate used 658,178 parameters compared with Hybrid's 1,521,672, but did not provide a sustained performance gain. All configurations and seeds, including the masking, feature-subspace and consistency experiments, are reported in Supplementary Tables S4–S5.

Simple linear controls also recovered discriminatory information from the fixed features. Logistic regression on mean features gave external AUROC 0.8335 on the common 512-patch reference inputs; averaging individually L2-normalized features gave 0.8462. Their sensitivities were 45.39% and 46.05%, with specificity 93.75% for both. Full-feature-mean inference yielded AUROCs of 0.8396 and 0.8475 (Supplementary Table S6). Ranking performance and fixed-threshold sensitivity therefore led to different comparisons.

Probability averaging across all five seeds and five new input bags produced external AUROC 0.8479 for Hybrid and 0.8314 for Attention+L2. The paired candidate-minus-Hybrid difference was −0.0164 (95% CI, −0.0448 to 0.0058; Supplementary Table S7). These ensemble AUROCs are distinct from

Table 3: Five-seed comparison of Hybrid and the internally selected candidate

Model	Validation AUROC	External AUROC	Ext. sens.	Ext. spec.
Hybrid	0.9348 ± 0.0134	0.8056 ± 0.1299	49.21%	82.50%
Attention+L2	0.9341 ± 0.0084	0.8061 ± 0.0916	55.66%	81.25%

AUROCs are means ± sample SD across seeds 17, 29, 43, 59 and 71; sensitivity and specificity are means of seed-specific rates at 0.5. These are not ensemble metrics. The same 49 validation and 168 external slides were reused. The three-seed selection was fixed before the additional seeds and new external inference.

the mean AUROC across separately trained models.

2.5 Input sampling affected individual decisions more than aggregate discrimination

For the retained Hybrid checkpoint, 20 additional patch samples per slide gave internal AUROCs of 0.9498–0.9682 and external AUROCs of 0.8549–0.8943 (Fig. 4). Median per-slide score standard deviations were 0.0120 internally and 0.0150 externally. Nevertheless, seven internal and 18 external slides crossed threshold 0.5 across the 20 draws. Averaging the scores yielded external AUROC 0.8775, with 141 true positives and 12 true negatives: one fewer false positive and one additional false negative than the retained single-bag result (Supplementary Table S8).

Model-selected regions were linked back to image windows for pathological review (Supplementary Fig. S4). The maps represent differences in similarity to two latent prototype banks. Their disease-specific morphological meaning has not been established. Performance and visualization runs used different sampled inputs; both scores are retained for the 12 illustrated cases, with no threshold classification changes in those cases (Supplementary Table S9). Region matching establishes where the displayed patches came from, rather than validating lesion localization.

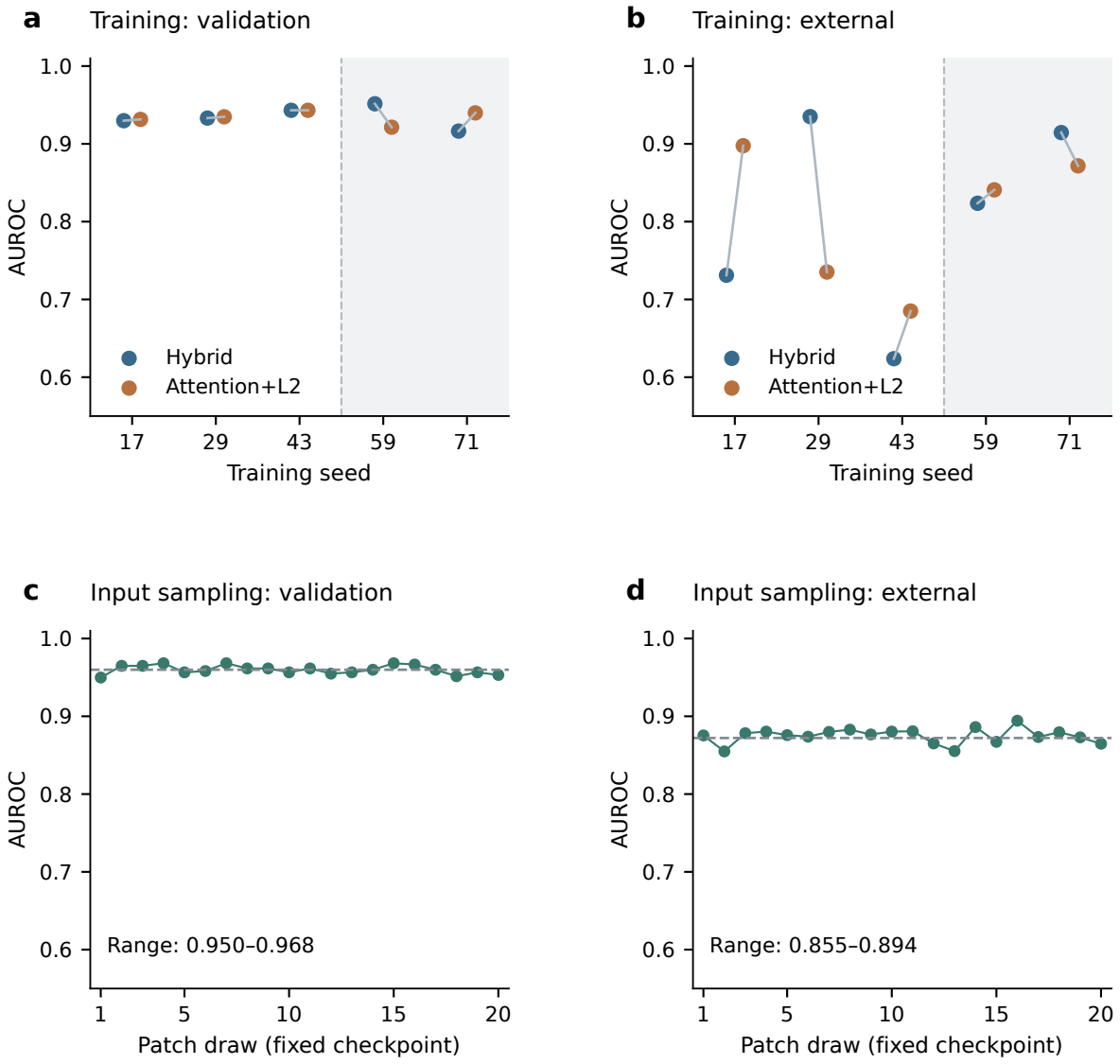


Figure 4: Training and input-sampling variation are different sources of uncertainty. a,b, AUROC for every training seed of Hybrid and Attention+L2 on internal validation and external evaluation. Gray shading identifies the two additional seeds after candidate selection on the first three internal results. Lines pair model results within each seed; seeds are not independent clinical samples. **c,d,** AUROC across 20 complete-cohort patch draws for the retained historical Hybrid checkpoint. Dashed lines show its original reference-bag AUROC. These are technical sampling distributions, not confidence intervals; the historical checkpoint differs from the retrained models in a,b. All cohort sizes remain 49 internally and 168 externally.

3 Discussion

This study found a MALT–SS classification signal in frozen pathology foundation-model features from salivary gland slides. The retained Hybrid model combined high internal validation discrimination with external AUROC 0.8721. The more consequential finding for model development was that this signal was also accessible to substantially simpler classifiers, while controlled experiments did not establish a stable advantage for the full aggregation architecture. Together, these results support further evaluation of the disease-specific task and provide a clearer basis for choosing which parts of the pipeline require improvement.

The clinical endpoint distinguishes this work from nearby applications. Automated SS classification and focus-score assessment quantify autoimmune gland pathology, whereas orbital MALT–IgG4 classification and lymphoma subtype classification address different tissues or differential diagnoses [5–8]. Our results extend image-based investigation to the recorded MALT/SS categories in salivary gland material, with whole-slide labels rather than manually annotated training regions. They do not establish performance specifically in diagnostically equivocal biopsies. Defining that case spectrum is important: success on an archive of established diagnoses need not imply the same benefit when morphology is borderline and ancillary testing is being considered.

The baseline results place the Hybrid model in an appropriate role. Its five summaries encode complementary mathematical descriptions of the sampled features, but representational variety alone does not demonstrate added diagnostic information. Matched attention with normalization used 56.75% fewer parameters and had an external five-seed mean AUROC close to Hybrid's. Linear mean-feature models also had competitive ranking performance. These observations are compatible with substantial task information already being present in the frozen encoder space; they do not isolate the causal contribution of the encoder, because no matched cross-encoder experiment was performed. Recent MIL work likewise emphasizes training strategy and sampling when translating foundation features into slide predictions [14]. The local feature-subspace and masking experiments here were adaptations, not full reproductions of nnMIL or ACMIL [15].

Training variation and patch-sampling variation had different practical consequences. Repeated inputs to the retained checkpoint gave a comparatively narrow external AUROC range, whereas separately trained models produced more dispersed external results despite similar internal validation scores. This pattern suggests that optimization or learned representations deserve attention alongside patch coverage. It does not identify a specific source of domain sensitivity, because initialization, sampled training inputs and optimization trajectories vary together across seeds. At the individual-slide level, input sampling still mattered: a subset of slides changed threshold classification across draws. Repeated-bag evaluation can expose this dependence, although averaging did not consistently improve the error balance.

External operating-point results also depended on the evaluated material. Removing IHC/special-stain archive tags changed sensitivity much more than the AUROC point estimate. That observation argues for reporting ranking and threshold metrics separately and for defining the intended clinical input before interpreting external performance. It cannot determine the effect of actual staining, because archive labels are not confirmed stain identities and the SS series lacks a resolved stain classification. The small external SS group further limits specificity precision. The low NPV at threshold 0.5 does not support a rule-out use, and a low Brier score in this highly imbalanced collection does not by itself establish calibration. A clinically relevant threshold would need to be selected on development data for a defined use and then evaluated on new material.

The regional outputs provide a route to pathological interrogation, rather than a demonstrated explanation of diagnosis. High- and low-margin patches are selected by relative similarity to latent prototype banks; those banks have no independent regional disease supervision. Similarly, a large

gate weight is a scaling coefficient within a learned fusion network, not a causal share of the prediction. Reviewing the selected patches alongside randomly sampled regions could test whether these internal signals capture relevant infiltrate composition or glandular context. Such evidence is needed before assigning disease-specific meaning to the maps or claiming reliable lesion localization.

Several boundaries determine the interpretation of this retrospective study. Internal validation was used for checkpoint selection. The external cohort was evaluated during successive development analyses, so the later algorithm comparisons are exploratory reuse of that cohort. The current numerical analyses use slides as their unit; patient linkage, clinical reference-standard details, collection periods and confirmed material categories remain to be incorporated from the hospital records. The fixed-prediction intervals do not account for patient clustering or model-selection uncertainty. Finally, this manuscript does not report a paired reader study or an AI-assisted reading experiment. Completing the clinical metadata and evaluating a frozen pipeline in a new, representative patient cohort would address a more consequential gap than additional architecture searches on the same external slides.

In conclusion, frozen pathology features supported retrospective salivary gland MALT–SS classification, but the evidence favored careful characterization of the learned signal over a claim of aggregation superiority. Simple baselines, controlled retraining and repeated inputs made the distinction between discrimination, stability and a clinical operating point explicit. These findings provide a practical foundation for a focused clinical validation study.

4 Methods

4.1 Study design and reference categories

This retrospective study evaluated WSI-level classification of recorded salivary gland MALT lymphoma and SS diagnoses. MALT was the positive class and SS the negative class. An SS medical history in a patient with MALT was a separate clinical attribute, not a reason to assign that slide to the SS class. Training used slide labels without cell-, patch- or lesion-level supervision. Clinical sources, dates, patient–specimen–slide linkage and case characteristics are reserved in Table 1. The source records identify Peking University School and Hospital of Stomatology for the internal material and Peking University Third Hospital for an external series; the complete provenance of the separate HE-directory material remains a field for the hospital records.

Reference standard. [H02: hospital-held diagnostic criteria, reference-review procedure, reviewer roles, ancillary tests and adjudication.] The exact criteria used in the source records will be stated here, rather than inferred from general lymphoma or SS classification guidance [3, 4, 16].

Eligibility and selection. [H03: clinical inclusion/exclusion rules, selection personnel and chronology, reasons for four prediction failures and 55 non-inclusions after successful prediction.] The recorded archive, successful inference and final analysis sets were retained as distinct stages. The analysis did not redefine cohort membership. Reporting was structured with reference to STARD-AI; unresolved items are listed in the accompanying completion checklist [17].

4.2 Image preparation and data partition

TRIDENT with GrandQC was used for tissue segmentation and patch extraction [18]. The 193 internal and 168 analyzed external WSI headers recorded a Hamamatsu C9600-12 scanner, a 40× source-lens setting and approximately 0.225708 μm per level-0 pixel. Non-overlapping 512×512-pixel patches were extracted at target 20× magnification, corresponding to $1,024 \times 1,024$ level-0 pixels. The frozen UNI2-h encoder generated 1,536-dimensional features [12, 19]. Its archived preprocessing

resized the shorter image side to 224 pixels, applied a 224×224 center crop and ImageNet channel normalization. [T01: exact encoder-weight checksum and downloaded revision.] Scanner-header details and the implementation record are provided in Supplementary Methods.

The 193-slide UNI2-h dataset was divided by label-stratified random sampling with seed 42 and a validation fraction of 0.25, yielding 144 training and 49 validation slides. The existing author-reviewed partition was reused in all controlled experiments. Each neural input contained 512 patch vectors, sampled without replacement when possible and with replacement otherwise. Training inputs varied across epochs or loading events; reference evaluation bags were deterministic. All reported counts and resampling units are WSIs. Patient-level quantities remain blank pending incorporation of the corresponding records.

4.3 Hybrid-Evidence MIL

Patch features were projected from 1,536 to 256 dimensions using a linear layer, layer normalization, GELU activation and dropout. The model constructed five vectors: gated attention pooling [9], the feature mean, the sample standard deviation, a high-margin feature mean, and the sum of a low-margin mean and coordinate-group summary. Two banks of eight learnable prototypes supplied maximum cosine similarities; their difference defined the margin. High- and low-margin summaries averaged the 32 extreme instances in each direction. Regional pooling operated on 16 equal-sized chunks after coordinate sorting, without an explicit anatomical-region annotation or adjacency graph.

Let $\mathbf{e}_1, \ldots, \mathbf{e}_5 \in \mathbb{R}^{256}$ denote these vectors and $\mathbf{u} \in \mathbb{R}^8$ the similarity/margin statistics. A learned gate g and fusion/classification network f produced

$$\boldsymbol{\gamma} = \operatorname{softmax}(g([\mathbf{e}_1; \cdots; \mathbf{e}_5; \mathbf{u}]))\,, \tag{1}$$

$$p = \operatorname{sigmoid}(f([\gamma_1 \mathbf{e}_1; \cdots; \gamma_5 \mathbf{e}_5; \mathbf{u}]))\,. \tag{2}$$

The statistics bypassed branch scaling. The full mathematical definitions, dimensions and operators are retained in Supplementary Methods. The score p is an uncalibrated MALT score. Training minimized weighted binary cross-entropy,

$$\mathcal{L} = -\frac{1}{B} \sum_{b=1}^{B} \left[\frac{77}{67} y_b \log p_b + (1 - y_b) \log(1 - p_b) \right], \tag{3}$$

implemented from logits. The class weight was the ratio of SS to MALT training slides. No region-level or prototype-specific objective was used in the original model.

4.4 Training and comparator analyses

The retained Hybrid model used AdamW, learning rate 5×10^{-5}, weight decay 10^{-4}, batch size eight, dropout 0.3 and 30 epochs. Seed 42 was used in the original experiment, and checkpoint selection by validation AUROC retained epoch 15. Validation performance was not interpreted as an independent test estimate. The original training environment has not been fully recovered (completion item T02).

Historical comparisons included mean pooling, ABMIL and local attention, transformer and prototype aggregators. Their architecture dimensions and training budgets differed. We use descriptive implementation names and preserve mappings to the archived code in Supplementary Table S1. These local variants are not complete reproductions of CLAM, DSMIL, DTFD-MIL, TransMIL or R^{2}T-MIL [11, 20–23]. CONCH v1.5 with TITAN used a different encoder and 1,024 instances

per slide and was treated as a separate pipeline [24]. The historical three-model ensemble and its development-selected threshold are reported only as exploratory supplementary analyses.

The matched comparison trained attention-only, full fixed-scaling and full learned-gate models with seeds 17, 29 and 43. Within each seed, shared parameters, sampled training bags and minibatch orders were matched. All runs used the Hybrid optimization settings above and the first strictly maximal validation-AUROC epoch. The models differed in parameter count; the comparison did not isolate capacity from gating.

The later algorithm screen used ten configurations and the same three seeds, followed by two additional seeds (59 and 71) for the internally selected candidate and Hybrid. The six control reruns reproduced the corresponding earlier matched-attention and Hybrid runs; they were not additional independent seeds. Candidate selection used three-seed mean validation AUROC, with the complete ranking frozen before the new external inference. The selected configuration remained unchanged after the additional seeds. Normalization, attention masking, feature-subspace aggregation and two-bag consistency settings are specified in Supplementary Methods and the archived protocol [14, 15, 25].

Two linear controls fitted StandardScaler and L2-regularized logistic regression only to the 144 training slides, using each slide's full-feature mean, with or without per-patch L2 normalization. Logistic settings were $C = 1$, balanced class weights, LBFGS, 3,000 maximum iterations and tolerance 10^{-8}, without tuning. Inference on 512-patch reference means and full-feature means was reported separately.

4.5 External evaluation, input sampling and regional outputs

The original external predictions used the retained Hybrid checkpoint and 512 patches sampled with seed 42 plus the original manifest index. Later ABMIL and two local comparator checkpoints were evaluated on exactly those reference inputs. Source and stain archive tags were retained without converting them to confirmed histological metadata. Omitting IHC/special-stain tags was a post hoc descriptive sensitivity analysis, not a revised primary cohort. Later algorithm evaluations reused the same historical external collection after it had already been observed during development.

For input-sampling analysis, the retained Hybrid checkpoint was evaluated on 20 additional deterministic bags per slide. We calculated per-slide score standard deviations, occurrence of both classifications around 0.5, and complete-cohort AUROC for each draw. Separately, all five seeds of Hybrid and Attention+L2 were evaluated on five new bags per slide. Seed and bag ensembles used arithmetic means of probabilities. Exact sampling rules and the distinct feature-window logit averaging used by the subspace model appear in Supplementary Methods. Repeated bags and seeds were not treated as additional clinical observations.

Regional displays reused archived margin maps and patch thumbnails without altering tissue pixels. Twelve visualization cases had their performance and interpretation scores linked, because the two scripts could sample different bags. Eight displayed image windows were matched to the archived crops. These checks support image traceability; pathological regional annotation and reader comparisons remain outside the current results.

4.6 Statistical analysis

The primary metric was AUROC. AP was calculated by the average-precision definition; threshold metrics used 0.5 unless a supplementary analysis explicitly stated otherwise. Sensitivity, specificity, predictive values, accuracy and confusion counts were calculated from aligned predictions. Balanced

accuracy was the mean of sensitivity and specificity. Brier scores were mean squared differences between scores and binary labels.

Primary fixed-checkpoint AUROC/AP intervals used 2,000 class-stratified WSI bootstrap samples with seed 20261016; proportion intervals used Wilson's method [26]. Historical internal paired AUROC intervals used aligned resampling; later external and algorithm comparisons used 10,000 class-stratified paired samples with seed 20261104. These conditional intervals do not incorporate checkpoint/configuration selection, patient clustering or training variation. Comparisons were descriptive and not multiplicity-adjusted. Across training seeds we report individual values and means with sample standard deviations, separately from AUROC calculated after probability averaging. Statistics and figures used Python; numerical source tables and executable rendering code accompany the draft.

Declarations

Ethics approval. The study protocol was approved by the Biomedical Ethics Committee of Peking University School and Hospital of Stomatology (PKUSSIRB-2025110092). The approval letter records review on 24 April 2025 and signature on 27 May 2025. [H04: consent or waiver wording, external-material coverage and applicable continuing-review provisions from the hospital records.]

Data availability. Publication-coded predictions, numerical source tables and analysis specifications accompany this draft. Raw WSIs, patient linkage and clinical records are not included in the draft source bundle. [A03: institution-approved data-sharing scope, repository/access route and identifiers.]

Code availability. The draft includes code for regenerating its quantitative figures and tables. Training and evaluation implementations, protocols and checkpoint identities are retained with the project records. [A04: author-controlled repository, release licence and trained-model access arrangement.] UNI2-h is obtained through its official model repository [19]; no foundation-model weights are redistributed here.

Author contributions. [A05: author-specific contributions.]

Funding and acknowledgements. [A06: funders, grant identifiers, funder roles and acknowledgements.]

Competing interests. [A07: all-author declaration.]

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